

Student Name

Gosford High School

2009

Higher School Certificate

Mathematics – Extension 1

Half-Yearly Examination - 2009

Time Allowed – 1 Hour 30 minutes
+ 5 minutes reading time

Remember to start each new question in a new booklet

Students must answer questions using a blue/black pen and/or a sharpened B or HB pencil.

Approved scientific calculators may be used

Students need to be aware that

- * 'bald' answers may not gain full marks.
- * untidy and/or poorly organised solutions may not gain full marks.

Marks

Question 1 Preliminary / 16	
Question 2 Integration / 16	
Question 3 Trigonometry / 16	
Question 4 Polynomials / 10 Circle Geometry / 5	
Total / 64	

Mathematics Extension 1 – Half Yearly Examination Year 12 – 2009**Question 1 (16 Marks)****Marks**

- a) Solve $\frac{1}{2x-1} \leq 2$ **3**
- b) P divides the interval from (4,-1) to (0,3) in the ratio k:1 **3**
(i) Write down the coordinates of P.
(ii) If P lies on $x+2y=0$, find the coordinates of P.
- c) Prove that, if $x^4 - x^3 + kx - 4$ has a factor of $(x+1)$, then it also has a factor of $(x-2)$. **3**
- d) When the polynomial $P(x)$ is divided by x^2-1 the remainder is $3x-1$.
What is the remainder when $P(x)$ is divided by $x-1$? Justify your answer. **2**
- e) (i) Show that the acute angle θ between the straight lines $y = x + 2$ and $y = mx + b$ is given by $\tan \theta = \left| \frac{m-1}{m+1} \right|$. **1**
- (ii) Write down a similar result for the angle ϕ between the straight lines $y = 3x - 1$ and $y = mx + b$. **1**
- (iii) Hence find the gradient(s) m of the line(s) $y = mx + b$ bisecting the angles between the straight lines $y = x + 2$ and $y = 3x - 1$.
(Answer in simplest surd form) **3**

Question 2 (16 Marks) Start a New Booklet**Marks**

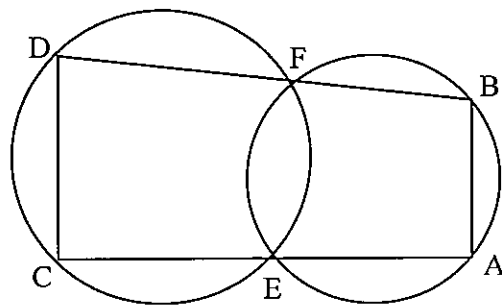
- a) Evaluate $\int_{\frac{1}{2}}^1 4t(2t-1)^5 dt$ by using the substitution $u = 2t - 1$ **3**
- b) Use the substitution $u = x^2 + 2$ to evaluate $\int_0^1 \frac{x}{(x^2+2)^2} dx$ **3**
- c) (i) Prove that $\int_0^{\frac{\pi}{4}} \sin^2 x dx = \frac{\pi}{8} - \frac{1}{4}$. **2**
- (ii) Find $\frac{d}{dx}(x \sin^2 x)$ **2**
- (iii) Hence, or otherwise, prove $\int_0^{\frac{\pi}{4}} x \sin 2x dx = \frac{1}{4}$. **2**
- d) (i) Divide x^3 by $x^2 + 1$ expressing your answer in the form $x^3 = (x^2 + 1)Q(x) + R(x)$. **2**
- (ii) Hence, or otherwise, evaluate $\int_0^1 \frac{x^3}{x^2+1} dx$ **2**

Question 3 (16 Marks) (Start a New Booklet)**Marks**

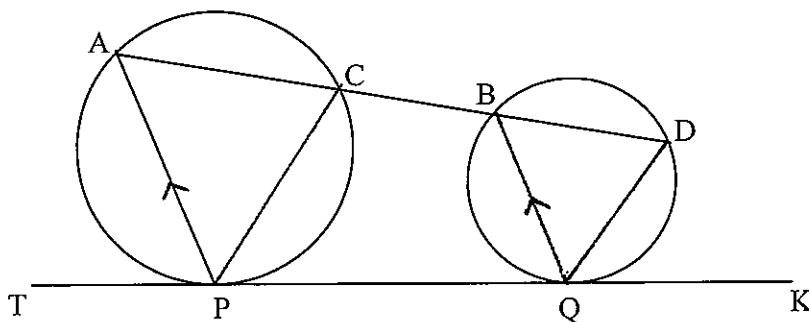
- a) Find in simplest exact form, the value of $\tan 105^\circ$ **2**
- b) Express $\cos x + \sqrt{3} \sin x$ in the form $A \cos(x - \alpha)$.
Hence solve $\cos x + \sqrt{3} \sin x = 1$ for $0 \leq x \leq 2\pi$. Give exact answers. **3**
- c) (i) Prove that $\sin(\alpha + \beta) + \sin(\alpha - \beta) = 2 \sin \alpha \cos \beta$. **1**
- (ii) Hence evaluate $\int_0^{\frac{\pi}{3}} \sin 2x \cos x \, dx$ **3**
- d) (i) Solve the equation $\sin 2x = \sin x$ for $0 \leq x \leq 2\pi$. **2**
- (ii) Give general solutions to this equation. **2**
- (iii) Show that if $0 < x < \frac{\pi}{3}$, then $\sin 2x > \sin x$. **1**
- (iv) Find the area enclosed between the curves $y = \sin 2x$ and $y = \sin x$ **2**
- for $0 \leq x \leq \frac{\pi}{3}$

Question 4 is on next page.

- a) The function $f(x) = x^3 - \ln(x+1)$ has one root between 0.5 and 1. 1
- (i) Show that the root lies between 0.8 and 0.9. 1
- (ii) Use halving the interval method to find the root correct to one decimal place. 1
- b) (i) Show that the function $f(x) = x^3 - x^2 - x - 1$ has a zero between 1 and 2. 1
- (ii) Taking $x = 2$ as a first approximation to this zero, use Newton's method to find a second approximation. 2
- (iii) Give a geometrical interpretation of the process used in (ii). Why is $x = 1$ unsuitable as a first approximation to this zero? 2
- c) The polynomial $(x - \alpha)^3 + \beta$ is zero at $x = 1$, and, when divided by x , the remainder is -7 . Find all possible values of the pair α and β . Hint: You may use the binomial expansion $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$. 3
- d) Two circles intersect at E and F. AEC and BFD are straight lines. Copy the diagram and prove that AB is parallel to CD. 2



- e) PQ is a common tangent to circles PAC and QBD
PA is parallel to QB.



- (i) Prove that PC is parallel to QD. 2
- (ii) PQBC is a cyclic quadrilateral. 1

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE:

$$\ln x = \log_e x, x > 0$$

Question 1

a) If $x > \frac{1}{2}$ solve If $x < \frac{1}{2}$ solve

$$1 \leq 4x - 2$$

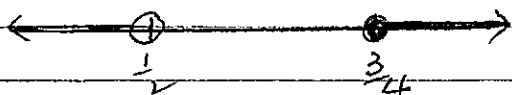
$$1 \geq 4x - 2$$

$$3 \leq 4x$$

$$3 \geq 4x$$

$$\frac{3}{4} \leq x$$

$$\frac{3}{4} \geq x$$



$$x < \frac{1}{2} \text{ or } x \geq \frac{3}{4}$$

b) i) $x = \frac{4}{k+1}$ $y = \frac{3k-1}{k+1}$

ii) $\frac{4}{k+1} + \frac{6k-2}{k+1} = 0$

$$6k + 2 = 0$$

$$k = -\frac{1}{3}$$

P is (6, -3)

c) $P(1) = 0 \therefore 1+1-k-4=0$
 $k = -2$

$$P(x) = x^4 - x^3 - 2x - 4$$

$$P(2) = 16 - 8 - 4 - 4 = 0$$

$\therefore x-2$ is a factor

d) $P(x) = (x^2-1)Q(x) + 3x-1$

$$P(1) = 0 + 3-1$$

$$P(1) = 2$$

remainder is 2.

e) For $y = x+2$, $m_2 = 1$
 For $y = mx+b$, $m_1 = m$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{m-1}{1+m} \right|$$

ii) For $y = 3x-1$, $m_2 = 3$
 $\tan \phi = \left| \frac{m-3}{1+3m} \right|$

iii) $\left| \frac{m-1}{m+1} \right| = \left| \frac{m-3}{3m+1} \right|$

$$|m-1||3m+1| = |m-3||m+1|$$

$$|3m^2 - 2m - 1| = |m^2 - 2m - 3|$$

$$\therefore 3m^2 - 2m - 1 = m^2 - 2m - 3$$

$$2m^2 + 2 = 0$$

No solutions

OR $3m^2 - 2m - 1 = -(m^2 - 2m - 3)$

$$4m^2 - 4m - 4 = 0$$

$$m^2 - m - 1 = 0$$

$$m = \frac{1 \pm \sqrt{1+4}}{2}$$

$$m = \frac{1 \pm \sqrt{5}}{2}$$

Question 2

a) $\int_0^1 4t(2t-1)^5 dt$

$$u = 2t-1$$

$$du = 2dt$$

$$t = \frac{1}{2} \quad u = 0$$

$$t = 1 \quad u = 1$$

$$= \int_0^1 4 \cdot \frac{(u+1)}{2} \cdot \frac{u^5}{2} du$$

$$= \int_0^1 u^6 + u^5 du$$

$$= \left[\frac{u^7}{7} + \frac{u^6}{6} \right]_0^1 = \frac{1}{7} + \frac{1}{6} = \frac{13}{42}$$

b) $\int_0^1 \frac{x dx}{(x^2+2)^2}$

$$u = x^2+2$$

$$du = 2x dx$$

$$x=0 \quad u=2$$

$$x=1 \quad u=3$$

$$= \int_2^3 \frac{du}{2u^2}$$

$$= \left[-\frac{1}{2u} \right]_2^3 = \left[\frac{-1}{2u} \right]_2^3$$

$$= -\frac{1}{6} + \frac{1}{4} = \frac{1}{12}$$

c) i) $\cos 2x = 1 - 2 \sin^2 x$

$$2 \sin^2 x = 1 - \cos 2x$$

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

2ci) cont.

$$\int_0^{\frac{\pi}{4}} \sin^2 x dx = \frac{1}{2} \int_0^{\frac{\pi}{4}} 1 - \cos 2x dx$$

$$= \left[\frac{x}{2} - \frac{\sin 2x}{4} \right]_0^{\frac{\pi}{4}}$$

$$= \left(\frac{\pi}{8} - \frac{1}{4} \right) - (0 - 0)$$

as required.

ii) $\frac{d}{dx}(x \sin^2 x) = x \cdot 2 \sin x \cos x + \sin^2 x$
 $= x \sin 2x + \sin^2 x$

$\therefore \int x \sin 2x + \sin^2 x dx = x \sin^2 x$

iii) $\int_0^{\frac{\pi}{4}} x \sin 2x dx = \left[x \sin^2 x \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \sin^2 x dx$
 $= \left(\frac{\pi}{4} \cdot \left(\frac{1}{2} \right)^2 - 0 \right) - \left(\frac{\pi}{8} - \frac{1}{4} \right)$

$= \frac{\pi}{8} - \frac{\pi}{8} + \frac{1}{4} = \frac{1}{4}$ as required

d) i)
$$\begin{array}{r} x \\ x^2+1 \overline{) x^3 + 0x^2 + 0x + 0} \\ \underline{-x^3} + x \\ -x \end{array}$$

$\therefore x^3 = (x^2+1)x - x$

ii) $\int_0^1 \frac{x^3}{x^2+1} dx = \int_0^1 x - \frac{x}{x^2+1} dx$
 $= \left[\frac{x^2}{2} - \frac{1}{2} \ln(x^2+1) \right]_0^1$

$= \left(\frac{1}{2} - \frac{1}{2} \ln 2 \right) - \left(0 - \frac{1}{2} \ln 1 \right)$
 $= \frac{1}{2} (1 - \ln 2)$

Question 3

a) $\tan 105^\circ = \tan(60^\circ + 45^\circ)$
 $= \frac{\tan 60^\circ + \tan 45^\circ}{1 - \tan 60^\circ \tan 45^\circ}$

$= \frac{\sqrt{3} + 1}{1 - \sqrt{3}} \times \frac{1 + \sqrt{3}}{1 + \sqrt{3}}$

$= -\frac{1}{2} (1 + 2\sqrt{3} + 3)$

$= -2 - \sqrt{3}$

b) $\cos x + \sqrt{3} \sin x = A \cos x \cos \alpha + A \sin x \sin \alpha$
 $A \cos \alpha = 1 \Rightarrow \tan \alpha = \sqrt{3}$
 $A \sin \alpha = \sqrt{3} \Rightarrow \alpha = \frac{\pi}{3}$
 $A^2 = 1^2 + (\sqrt{3})^2 \Rightarrow A = 2$

$2 \cos(x - \frac{\pi}{3}) = 1$

$\cos(x - \frac{\pi}{3}) = \frac{1}{2}$

$x - \frac{\pi}{3} = -\frac{\pi}{3} \text{ or } \frac{\pi}{3} \text{ or } 2\pi - \frac{\pi}{3}$

$x = 0 \text{ or } \frac{2\pi}{3}$

c) i) $\sin(\alpha + \beta) + \sin(\alpha - \beta)$
 $= \sin \alpha \cos \beta + \cos \alpha \sin \beta + \sin \alpha \cos \beta - \cos \alpha \sin \beta$
 $= 2 \sin \alpha \cos \beta$ as required

ii) Put $\alpha = 2x$ + $\beta = x$

$$\int_0^{\frac{\pi}{3}} \sin 2x \cos x dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{3}} \sin 3x + \sin x dx$$

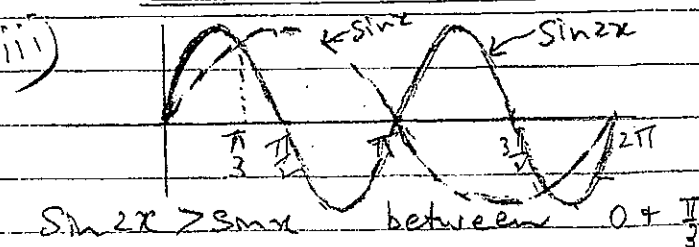
$$= \frac{1}{2} \left[-\frac{1}{3} \cos 3x - \cos x \right]_0^{\frac{\pi}{3}}$$

$$= \frac{1}{2} \left\{ \left(-\frac{1}{3} \cdot -1 - \frac{1}{2} \right) - \left(-\frac{1}{3} - 1 \right) \right\}$$

$$= \frac{1}{2} \left\{ -\frac{1}{6} + \frac{4}{3} \right\} = \frac{7}{12}$$

d) i) $2 \sin x \cos x - \sin x = 0$
 $\sin x (2 \cos x - 1) = 0$
 $\sin x = 0 \text{ or } \cos x = \frac{1}{2}$
 $x = 0, \pi, 2\pi \quad x = \frac{\pi}{3}, \frac{5\pi}{3}$

ii) $x = n\pi \text{ or } x = 2n\pi + \frac{\pi}{3}$



$$\begin{aligned}
 \text{iii) Area} &= \int_0^{\frac{\pi}{3}} \sin 2x - \sin x \, dx \\
 &= \left[-\frac{\cos 2x}{2} + \cos x \right]_0^{\frac{\pi}{3}} \\
 &= \left(-\frac{1}{2} \cos \frac{2\pi}{3} + \cos \frac{\pi}{3} \right) - \left(-\frac{1}{2} \cos 0 + \cos 0 \right) \\
 &= \frac{1}{4} + \frac{1}{2} + \frac{1}{2} - 1 \\
 &= \frac{1}{4} \quad \underline{\underline{u^2}}
 \end{aligned}$$

Question 4

$$a) f(x) = x^3 - \ln(x+1)$$

$$i) f(0.8) = .8^3 - \ln 1.8 \\ \approx -0.0758$$

$$f(0.9) = .9^3 - \ln 1.9 \\ \approx 0.0871$$

Function is defined & continuous in this region
 $\therefore$ Root lies between 0.8 & 0.9

$$ii) f(0.85) = .85^3 - \ln 1.85 \\ \approx -0.00106$$

Root lies between 0.85 & 0.9. $\therefore$ To one decimal place the root is 0.9.

$$b) i) f(1) = 1 - 1 - 1 - 1 = -2 \\ f(2) = 8 - 4 - 2 - 1 = 1$$

Function is continuous & defined in the region.

One positive, one negative. Root lies between 1 & 2.

$$ii) x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$f'(x) = 3x^2 - 2x - 1$$

$$f'(2) = 12 - 4 - 1 = 7$$

$$x_2 = 2 - \frac{1}{7}$$

$$x_2 = 1\frac{6}{7} \approx 1.857$$

iii) The tangent at $x=2$ cuts x axis at $1\frac{5}{6}$. This is a better approximation provided there is no stationary point in the vicinity. $f'(1) = 0$
 $\therefore$ Stationary at $x=1$
 $(x_2 = 1 - \frac{2}{0} \text{ undefined})$

$$\begin{aligned}
 c) P(x) &= (x-2)^3 + \beta \\
 P(1) &= 0 = (1-2)^3 + \beta \quad \text{--- (1)} \\
 P(0) &= -7 = (-2)^3 + \beta \quad \text{--- (2)} \\
 1 - 3x + 3x^2 - x^3 + \beta &= 0 \\
 1 - 3x + 3x^2 - 7 &= 0 \\
 3x^2 - 3x - 6 &= 0 \\
 (x-2)(x+1) &= 0 \\
 x = 2 \text{ or } -1 \\
 -\beta = 1 \text{ or } -8
 \end{aligned}$$

d) Join EF
 $\angle DFE = \angle BAE$ (exterior angle of cyclic quad ABFE)
 $\angle DCE = 180^\circ - \angle DFE$
 (Opposite angles of cyclic quad are supplementary)
 $\therefore \angle DCE + \angle BAE = 180^\circ$
 But these are interior $\therefore AB \parallel CD$

e) $\angle APT = \angle ACP$ } Angle between tangent & chord
 $\angle BQP = \angle BDQ$ } equals angle in alternate segment
 But $\angle APT = \angle BQP$
 Corresp $\angle$'s $AP \parallel BQ$
 $\therefore \angle ACP = \angle BDQ$. But these are corresponding $\therefore PC \parallel DQ$

ii) $\angle APT = \angle ACP$ (alt segment)
 $\angle APT = \angle BQP$ (corresp $\angle$'s)
 $AP \parallel BQ$
 $\therefore \angle ACP = \angle BQP$. But this is exterior $\angle \therefore PQBC$ is cyclic.